

AAROHAN 2026

ASCENT OF KNOWLEDGE

Class 10th

Name

Contact Number

Instructions: -

1. This Question Paper contains 4 comprehensive passages based on the above theme. Each passage is followed by some questions. There will be 24 questions in all.
2. Unless otherwise stated, each question has **ONLY ONE** out of the four choices as the correct answer. Marking Scheme -> +4 for correct Answer, -1 for incorrect Answer and 0 for not attempting the question.
3. Questions marked as MAQ are Multiple Answer type questions. +4 will be awarded when only all correct options are marked. 0 for not attempting and -1 in all other cases.
4. The questions will test your skills in Physics, Chemistry, Mathematics and Analytical Reasoning, along with your understanding of the passage.

ALL THE BEST (ノ°-°)ノ

ENTANGLED SECRETS

In quantum physics, **entanglement** creates a powerful link between particles, such as **photons**, so that their properties remain **strongly correlated**, even when separated by vast distances.

This unusual feature is the foundation of **quantum communication systems**, where information is encoded in the **states of photons**.

Quantum cryptography, particularly protocols like **BB84**, relies on the principle that measuring a quantum particle disturbs it. Each photon is prepared and measured in one of two possible bases:

- **Rectilinear basis** (+) → vertical (|) or horizontal (—) polarization
- **Diagonal basis** (×) → 45° or 135° polarization

When both the sender and receiver use the **same basis**, their results should **match perfectly**. If they use **different bases**, the results appear **random** and are later discarded during the key sifting process.

If an eavesdropper tries to intercept the photons, the act of measurement alters their states, introducing detectable errors. This property makes quantum cryptography fundamentally different from classical encryption, where interception might go unnoticed.

Despite its promise, there are several practical obstacles to implementing quantum communication systems:

- **Fragility of Entangled Photons:**
Entangled photons are delicate, and over long distances they suffer from decoherence due to scattering, absorption, or environmental noise, which destroys their correlations.
- **Transmission Limitations:**
Both optical fibers and free-space channels cannot maintain entanglement indefinitely.

- **Quantum Repeaters:**

Devices called quantum repeaters are being developed to extend secure communication links by re-establishing entanglement without directly measuring the photons.

- **Photon Source Reliability:**

Generating reliable streams of entangled photons is challenging and often requires advanced techniques such as nonlinear crystals or quantum dots.

Recent breakthroughs, like **China's Micius satellite mission**, have shown that secure quantum communication across thousands of kilometers is possible.

However, achieving a global quantum internet still faces significant technological and engineering challenges, including:

- Photon loss over long distances
- Maintaining synchronization between distant stations
- Improving photon source efficiency and stability

Quantum communication and cryptography offer unprecedented security based on the laws of physics, not computational difficulty.

Yet, to make this technology practical on a global scale, scientists must overcome the hurdles of **photon loss**, **decoherence**, and **entanglement preservation** across long distances.

1. A research team tries to build a global quantum internet using satellites. Which combination of challenges will they most likely face?

- A. Photon loss in space, weak entangled photon sources, need for quantum repeaters
- B. Photon overheating, laser instability, lack of vacuum in space
- C. Measuring photons too quickly, limited satellite battery, low temperatures
- D. Photon mass increasing with distance, electron loss, unstable encryption codes

2. The Micius satellite demonstrated long-distance quantum communication. What was the main significance of this achievement?

- A. It proved photons can travel faster than light in space
- B. It showed entanglement can survive across thousands of kilometers
- C. It allowed communication without using any ground equipment
- D. It replaced the need for quantum repeaters

3. In a quantum communication test, Alice and Bob find that 20% of their results are mismatched even when using the same basis. Which is the least likely explanation?

- A. An eavesdropper is tampering with the photons
- B. Environmental noise is causing decoherence
- C. Their detectors have calibration errors
- D. Entanglement allows random mismatches by design

4. Suppose in BB84, Alice uses entangled photons while Bob uses a laser to measure them. Why would this setup fail?

- A. Lasers produce light of only one color
- B. Lasers are too weak to measure photons
- C. Measuring with the wrong basis destroys the key correlations
- D. Lasers add extra photons, breaking the security

5. Imagine a bank is considering quantum cryptography for secure transactions. Based on challenges described, which factor would most limit its use in a crowded city environment?

- A. Fiber optic cables scatter and absorb photons
- B. Lasers are already in use for ATMs
- C. Electrons move faster in metals than photons in fibers
- D. Banking transactions cannot be encrypted with photons

6. Two entangled photons are sent to City A and City B. The difference in arrival time is recorded:

Channel	Fiber	Satellite
Avg. Delay (ns)	0.060	0.002

Which conclusion is most consistent with the data?

- A. Fiber links are faster than satellites but less synchronized
- B. Satellites maintain entanglement better because timing mismatches are smaller
- C. Entanglement disappears more quickly in satellites
- D. Photons in fiber do not experience decoherence

7. Alice and Bob share results for 8 photon pairs:

Basis Match?	Match?	Basis Match?	Match?
Yes	✓	Yes	✓
Yes	✗	No	Random
No	Random	Yes	✗
Yes	✓	Yes	✓

If normally all same-basis pairs should match, what percentage of same-basis results appear disturbed?

- A. 20%
- B. 25%
- C. 33%
- D. 40%

Space for Rough Work

SMALLER THAN YOU CAN SEE

On a clear night, you might gaze up and see countless stars, each a glorious pinprick of light. You might look down and see billions of humans, bustling with countless inventions. Yet, between these two extremes, it's easy to miss a secret world hiding in plain sight, a world filled with organisms more numerous than the stars and more diverse than humans themselves.

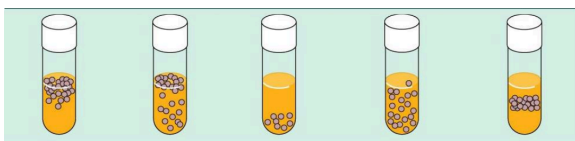
This world is home to microbes-creatures so tiny that an entire city of them could fit on the head of a pin. They are but a whisper away from the atomic world, yet they are living, breathing organisms with their own secrets.

Microbes play an irreplaceable role in our daily lives. Yet we hardly seem to notice them. Let's delve into the details of these discrete dwellers.

Firstly, just like you or me, these microbes require some essentials to exist, to grow and to reproduce. However, very unlike you or me, they may not require the same things as us. For example-Oxygen. Many kinds of bacteria have varied attitudes towards oxygen. Some love it as we do, while others hate it to the point that they cannot survive in its presence. We classify these bacteria based on their oxygen affinity as-

- i) **Aerotolerant Anaerobes**: Only anaerobic growth, but can survive in contact with oxygen.
- ii) **Obligate Anaerobes**: Only anaerobic growth, are harmed by oxygen presence.
- iii) **Obligate Aerobes**: Only aerobic growth, necessarily need oxygen to survive
- iv) **MicroAerophiles**: Only aerobic growth, but cannot handle even atmospheric oxygen levels
- v) **Facultative Anaerobes**: Grow best in oxygen, but can grow even in its absence at a lower rate

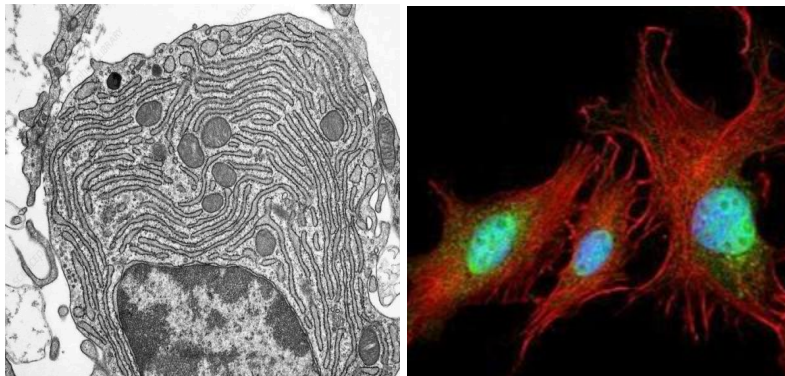
Equal amounts of each of these types of bacteria have been suspended in a solution and placed in normal aerobic conditions in the given experimental setup below. Try and identify which is which.



Let us label these test tubes from right to left, number 1 to number 5.

Space for Rough Work

Microbes derive their name from the Greek mikros(meaning small) and bios(meaning life). The mikros part of their name makes it impossible for us to view them with our naked eye. We use microscopes and various imaging techniques to view these interesting life forms. However, most microbes are nearly transparent and hence a simple light microscope gives very poor images. We use a combination of staining and microscope modification to obtain really clear and definite images such as the ones shown below.



Of the many methods that exist to obtain such images, we can focus on two commonly used ones- Gram Staining and Immunofluorescent staining.

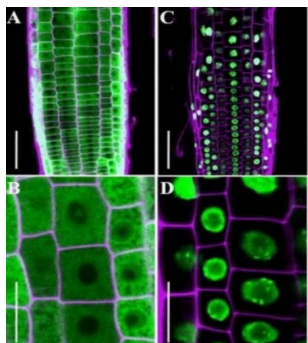
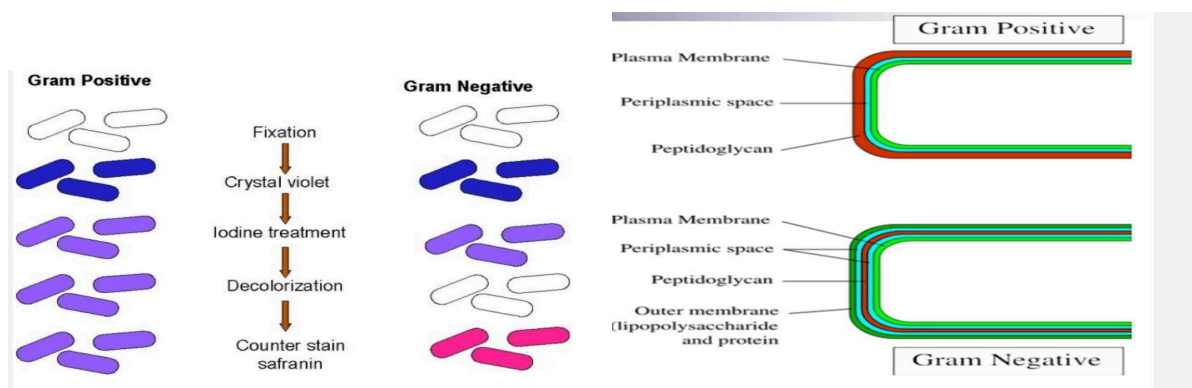
Gram staining is one of the most common staining methods out there. It's used to differentiate between Gram positive and Gram negative bacteria. It's a differential stain, so the two types of bacteria will end up stained two different colours, like this:



Space for Rough Work

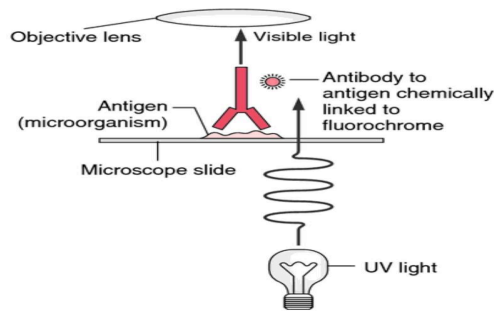
Gram staining involves four reagents applied in this order:

- i) **Crystal Violet** (dyes the bacterium violet)- Primary dye that enters the cell, permeating the cytoplasm.
 - ii) **Iodine**- Acts as a mordant. Meaning it forms large crystals with the primary dye that are too large to escape out through the cell wall.
 - iii) **Alcohol**- Decolorizing Agent. It washes away any dye it comes into contact with, but also dehydrates and shrinks the cell wall pores of any peptidoglycan cell walls it comes into contact with. This helps keep large dye crystals within the cell.
 - iv) **Safranin** (dyes the bacterium pink)- Counter Stain. Stains any colourless bacteria in another colour to help differentiate between the two types of cells.
- Crystal Violet is the dominant dye as compared to Safranin.



Immunofluorescent dying is a more modern method. It's a highly targeted way of dying cell components. It involves producing antibodies that target exactly the molecule/protein under interest. Then, a fluorescent dye molecule is attached chemically to the antibody. Hence, the dye gets carried by the antibody with high accuracy to the specific cell component under observation. It can produce really colourful and vivid images, such as these produced in BITS Pilani's own labs:

Space for Rough Work



In cases where we wish for higher resolution, we use electron microscopes instead of light microscopes. These use beams of electrons instead of light. In this case, we must use metal atoms instead of fluorescent molecules to attach to antibodies. The electron beams will be repelled by the electrons of the atom, creating a visible dark spot. The more the repulsion, darker the spot, more detail can be seen in the image.

Now that we know how to view them and under what conditions they grow, let's see how they multiply. Have you ever heard the phrase "breed like rabbits"? Well, bacteria breed much faster than rabbits. The most common method of bacterial reproduction is binary fission. This is a process in which one bacteria parent nucleus divides into two identical daughter cells in a set amount of time. After an equal amount of time, the two daughter cells split into two further daughter cells each, resulting in four cells. This process continues until nutrients run out.

1.If the entire experimental setup of test tubes and microbes is placed in an airtight space within which another experiment is also occurring-the complete combustion of propane.Now based on these new conditions,analyze which test tubes will have the most living cells post some time elapsing.

- A. I>II>III>IV>V
- B. III>IV>II>V>I
- C. IV>V>I>II>III
- D. II>I>V>IV>III

2.Let us repeat the same setup as Q1 with one small change. Instead of combustion of propane, we have electrolysis of water. Now what will the order of most living cells post some time elapsing be?

- A. I>II>V>IV>III
- B. II>III>I>V>IV
- C. IV>V>I>II>III
- D. I>III>V>IV>II

3.A careless student once made the mistake of entirely skipping the use of Iodine in his Gram staining process. What will be the final colour of Gram positive and Gram negative bacteria respectively for this careless student?

- A. Pink, Pink
- B. Purple, Pink
- C. Pink, Purple
- D. Purple, Purple

4.If you must choose a metal found in a typical Indian household for tagging an antibody in B. Satellites maintain entanglement better because timing mismatches are smaller

Explanation:

The smaller timing difference in the satellite link suggests better synchronization and less decoherence, helping maintain stronger entanglement correlations.

electron microscope immunofluorescent microscopy, which one would it be?

- A. Gold from Jewellery
- B. Aluminium from pans.
- C. Copper from wiring
- D. Iron from heavy kadai/tawa

5.In continuation of Q5,if I wish for my antibodies to attach themselves to two proteins which are 15 nm apart, what diameter of metal particles should I use?

- A. 30 nm
- B. 15 nm
- C. 10 nm
- D. 45 nm

6.Regarding bacterial growth,if I placed one bacterial cell in a test tube containing an excess of required nutrients at 10:00 AM. This bacterial cell divides every 20 mins and manages to fill the test tube with bacterial cells by 4:00 PM. During this process, how much time did the bacterial cell take to fill exactly 1/4th of the test tube?

- A. 1 hour 30 minutes

Space for Rough Work

- B. 3 hours
- C. 5 hours 20 minutes
- D. 4 hours 30 minutes

ENGINES BENEATH

THE MANTLE

The mantle, as you know, is a layer of earth's interior and it can be viewed as a kind of heat engine, powered by internal heat sources and cooled at the surface. One of the sources of heat is the radioactive decay of elements like uranium, thorium and potassium (another major source is residual primordial heat - yes "leftovers" from the earth's creation). These elements act as a "reservoir" while the earth's surface acts as a "sink". The mantle is the part of the said "engine" that converts this heat energy to physical "work" in the form of shell motion (tectonics and convection). As expected, energy conservation holds and a stable equation looks like:

$Q_{in} = Q_{rad} + Q_{prim} = Q_{out}$, where each symbol has its obvious meaning and Q denotes heat flow.

HEAT TRANSFER:MECHANISM

In the rigid crust (lithosphere), the main heat flow is by conduction which is flow due to direct physical contact. It is governed by Fourier's law of heat conduction, which mathematically is

$Q = -kA\Delta T \equiv q = -k\frac{dT}{dx}$ where Q is the heat flow per unit time, A is area of cross section in contact, k is the thermal conductivity and ΔT is temperature gradient.

However, the mantle is majorly made of highly viscous fluid, thus enabling thermal convection to be the dominant heat transport mechanism. To assess when convection is possible, we use a number called the Rayleigh number. Mathematically:

Space for Rough Work

$Ra = \frac{\rho \alpha g \Delta T d^3}{\kappa \eta}$ where κ is the thermal diffusivity and η is the viscosity. This Ra is a critical value, below which conduction is the major source of heat transfer and above which it is convection. Due to the usually high η of convection-able mantle, this can be scaled and the practical estimate of surface heat flux is scaled as:

$q_{surf} \sim k \frac{\Delta T}{\gamma_T}$ where γ_T is thermal boundary layer thickness.

RADIOACTIVE DECAY

As said before, this is a major source of heat in the interior of our planet. This is essentially due to long-lived isotopes (U^{238} , U^{235} , Th^{232} , K^{40}). Over time, these isotopes decay exponentially (concept related to half lives) which produce radiation and hence, heat.

CORE-MANTLE HEAT TRANSFER & GEODYNAMO

Some of the mantle's heat flow is derived from below, i.e., from the boundary of the outer core. This heat gradient drives the convection current in the liquid outer core region and this is where the geodynamo operates. In magnetohydrodynamics, we have an induction equation that can relate magnetic field, fluid velocity in the core and magnetic diffusivity.

For a rough idea, you need to know that geodynamo is the natural process that generates the magnetic field of the earth by convection of magnetic flux (molten iron and nickel).

The rate of heat loss of the core affects the convection currents which in turn affects magnetic field strength.

TECTONICS AND VOLCANOES AND THE MAGNETIC FIELD

The mantle's convection currents impose stresses and strains on the overlying lithospheric plates, driving phenomena like sea floor division, continental drifts, formation of mountains and associated volcanic activity as well. Hot ascending "plumes" can puncture the crust to form volcanoes and geothermal spots.

The magnetic field generated by the core acts as a shield against solar flares and cosmic radiation by deflecting off solar wind particles. If the geodynamo weakens, the field strength decreases, increasing risk to radiation fluxes, which can completely change the ecosystem as we know it.

1. With respect to the Rayleigh number's formula, which quantities increment would lead to high convection if other parameters are somehow kept constant. OR argue why increasing ΔT is not enough to promote more convection.

- A. Raising T always strengthens convection, since Ra is directly proportional to it; viscosity and other effects remain constant as temperature increases
- B. A larger ΔT increases buoyancy but simultaneously reduces viscosity with temperature that may inhibit large-scale convection
- C. Convection only depends only on ΔT and g ; other parameters will only lead to conduction
- D. If ΔT increases, it becomes easier to conduct the heat, which evens it out, keeping Ra constant

2. To sustain a dynamo, what would be more necessary to take into account? Actual transport of the magnetic materials in bulk (advection) or just the movement of these materials to create a field-like effect (diffusion)? For either case, what should the magnetic Reynolds number be to support your answer? The magnetic Reynolds number is a parameter that can be related to the flow state Reynolds number, theoretically giving the same useful information about current flow.

- A. Magnetic diffusion should dominate ($Ri \ll 1$) because the local magnetic deformities can homogenise to make an overall field

Space for Rough Work

- B. Advection should dominate ($RI \gg 1$) to carry the magnetic fluid in bulk to make an overall field
- C. Diffusion should dominate ($RI \gg 1$) since advection cannot make a permanent field; it would transfer fluids in bulk but it cannot be a constant generation which is required
- D. Advection should dominate ($RI \ll 1$) since diffusion is a random process and it cannot account for a continuous field

3. Why do tectonically active zones have large geothermal gradients and what information does that give us about q_{surf} ?

- A. Active zones have higher thermal surface boundary, increasing heat flow resistance
- B. Increase in mantle convection intensity leads to more q_{surf} leading to larger ΔT
- C. The q_{surf} is high because of the naturally occurring low lithosphere width (gamma in the formula) which facilitates larger ΔT
- D. The geothermal gradient is steep due to smaller q_{surf} leading to a lower ΔT which enhances tectonic activity

4. [MAQ] Assume the mantle as a proper heat engine. Let's say the total outgoing heat from mantle to surface and beyond is some value Q_{out} . Only a fraction of this drives convection and rest is lost by radiation. Let this fraction be f . If the convective region moves avg x cm/year against a pressure P , what is the effective efficiency of the mantle? You can assume that we call it efficient if all of the energy produced by the mantle is used to cause plate movement.

- A. The total heat loss by radiation is Qf
- B. The net heat used for plate movement is only $P \Delta x \cdot 315360$
- C. The efficiency is guaranteed to be less than 100% due to the land coverage area of the earth over the oceans

Space for Rough Work

D. All of the above

5. A mantle plume rises through 2900 km of mantle with average density d (4500 kg/m^3) and thermal coefficient of expansion k (0.00003 K^{-1}) and temperature excess ΔT (250 K). Estimate fractional density difference between plume and surroundings and thus get initial buoyant acceleration. (Assume that gravity remains constant throughout the interiority of the earth, $g = 10 \text{ m/s}^2$)

- A. 0.75% and 0.075 m/s^2
- B. 7.5% and 0.75 m/s^2
- C. 75% and 7.5 m/s^2
- D. 0.75% and 0.0075 m/s^2

6. Assume a hypothetical scenario where the earth ran out of its primordial heat but somehow still has long half-life isotopes in its composition, thus the total heat internal heat source is just a function of time exponentially. Let us assume that this heat source has a function $H(t) = H_0 e^{-kt}$ where t denotes time and e is a universal constant. Let the loss of heat from the surface be a constant value H_I . ($H_I < H_0$). Also assume that tectonic vigour is proportional to the square root of $(H - H_I)$ [obviously valid only when $H > H_I$]. What is the nature of plate activity and when will it stop? What does this mean geologically?

- A. Plate activity is a decreasing function of time. The heat has nowhere else to go which directly leads to surface temperature increment
- B. Plate activity stops abruptly which causes a sudden change in momentum. This destroys everything on the surface due to the immediate jerk

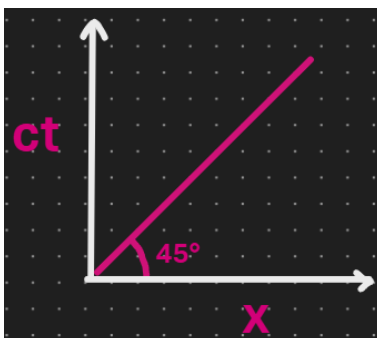
Space for Rough Work

- C. Plate activity stops as a smooth function, just when $H=H_I$. However, the surface temperature will decrease with time and the planet eventually dies
- D. Plate activity doesn't stop, in fact it stabilizes, which becomes nearly constant at infinite time. This causes an eventual constancy in surface temperature

TAMING TIME AND SPACE

Minkowski diagram

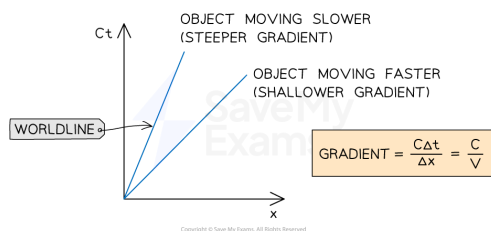
A spacetime (or Minkowski) diagram is basically a fancy way of drawing how things move through space *and* time at the same time. Since we obviously can't draw all four dimensions (x, y, z, and time) on paper, we cheat a little: we squash all of space into one horizontal line (x), and we put time on the vertical axis—but instead of just t, we use ct (which is just time multiplied by the speed of light). That way, both axes have the same units, so everything lines up nicely. So light moving through space would look something like:



Space for Rough Work

Now, when you draw something moving on this diagram, its path is called a worldline. The slope of that worldline tells you how fast it's moving:

45° diagonal line represents light itself — nothing can be flatter than this, because nothing beats light. (according to albert einstein's special theory of relativity, light is the cosmic speed limit)



What's cool is that this one diagram helps you see things like time dilation, length contraction (Motion causes lengths to shrink along the line of travel) and simultaneity —all the weird stuff from relativity suddenly becomes a picture instead of equations. Even causality (what events can affect each other) pops out naturally: anything outside the light line is basically “too far too fast” to influence you.

An event and Light cones

In a spacetime diagram, an event is just one exact point where something happens — meaning it has:

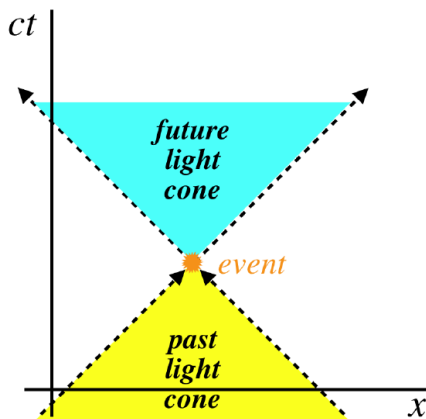
- One position in space (a specific x -value)
- One moment in time (a specific t -value)

So on the graph, an event is simply a single dot with coordinates (x, ct) .

For example, “a torch is switched on at $x = 3$ meters at time $t = 2$ seconds” — that's one event → one dot on the diagram.

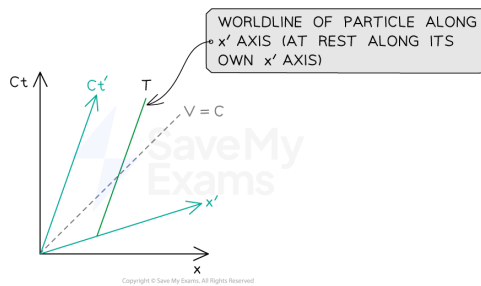
In a spacetime diagram, every event—like a flash of light or someone dropping a ball—gets its own

light cone. From that event, you draw two diagonal lines at 45° (because that's the path light would take going left and right at speed c). Everything inside the upper cone is what *can be influenced* by that event in the future. Everything inside the lower cone is what *could have influenced* it in the past. Anything *outside* the cone is completely unreachable—you could scream, send rockets, or fire lasers, but nothing gets there in time.



Lorentz transformation

A Lorentz transformation shows how two observers moving relative to each other describe space and time differently. In the diagram, the black axes (x and ct) are for an observer at rest, while the blue axes (x' and ct') are for an observer moving at a constant speed. The moving observer's time axis tilts toward the direction of motion, and the space axis tilts toward the light line, so that both observers still measure the speed of light as the same. This tilt reflects that motion mixes space and time: what one observer sees as “pure space” or “pure time” is a combination of both for someone moving.

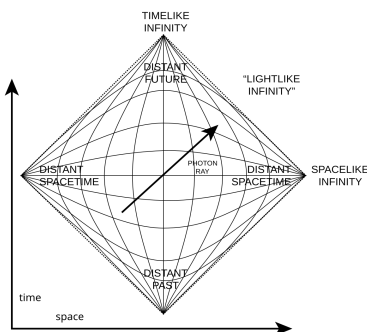


Penrose diagrams

Minkowski diagrams are great for showing motion, worldlines, and light cones, but they have one big limitation:

The graph is infinite. Light rays go off forever, objects moving forward in time travel endlessly up the diagram, and distant regions can't be shown properly.

To fix this, physicists came up with a clever trick: Squish the entire infinite spacetime into a finite box - without breaking light cones. And that's called a Penrose Diagram (or *conformal diagram*).

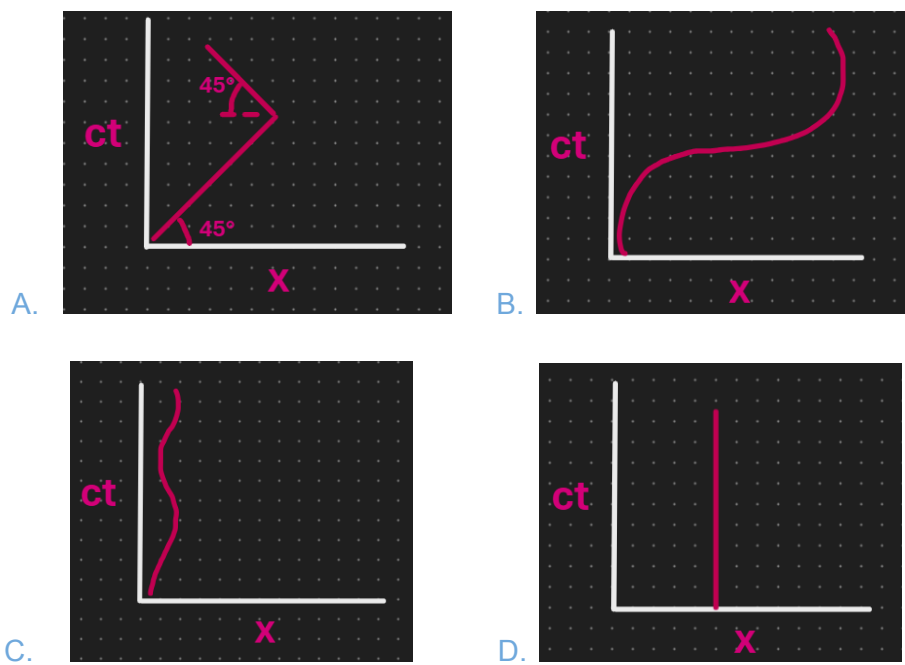


Black Holes:

Imagine the black hole as a "trap" in spacetime. On a Penrose diagram, the event horizon shows up as a diagonal line. Once you cross it, all possible paths (worldlines) tilt inward toward the singularity — even light can't escape. (Value of $c = 3 \times 10^8$ m/s)

Space for Rough Work

1. Which of the following worldlines is not possible?



Space for Rough Work

2. An event E occurs at the origin ($x = 0$, $t = 0$) on a spacetime diagram (x horizontal, ct vertical). Light travels at speed $c = 3 \times 10^8$ m/s. Another event F occurs at $x = 2 \times 10^8$ m and $t = 1$ s. What is true about the events E and F ?

- A. The slope is 0.67, and F can be influenced by E.
- B. The slope is 1.5, and F cannot be influenced by E.
- C. The slope is 1.5, and F can be influenced by E.
- D. The slope is 0.67, F cannot be influenced by E.

3. Quantum entanglement relation with spacetime

A physicist at point P makes a measurement on one half of an entangled pair. The other half is rushing toward point Q, which is space-like separated from P. The physicist makes the following independent arguments:

- i) "Since the wavefunction collapse is instantaneous, my measurement at P causes the state at Q to be definite, violating the light cone principle."
- ii) "The correlation pattern between the two particles was established at the initial entanglement event, which strictly lies in the past light cone of both P and Q."
- iii) "An observer at Q cannot verify that a "collapse" occurred due to P until they receive a light-speed signal from P, ensuring no FTL information transfer."
- iv) "For the wavefunction description to be frame-independent, the measurement events P and Q must be connected by a worldline of influence that lies inside a light cone, meaning Q must be in the future light cone of P for the theory to be consistent."
- v) "In some inertial frames, the measurement at Q occurs before the measurement at P. In those frames, it is the measurement at Q that causes the collapse at P, proving the influence is frame-dependent and thus not a true causal violation."

The wavefunction describes the state of a quantum particle and tells us the probability of finding it in different places.

Which of the following arguments made by the physicist are incorrect in context of relativistic causality?

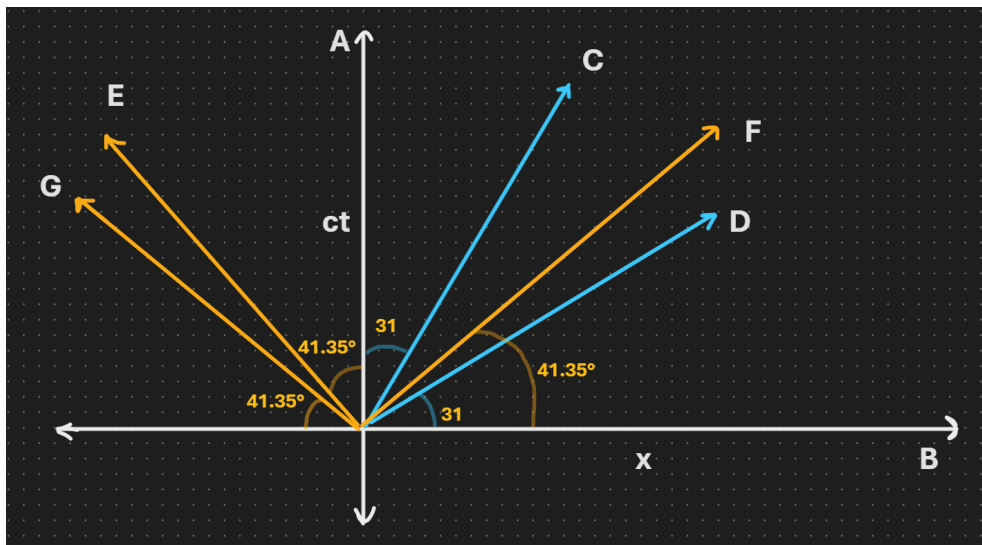
Space for Rough Work

- A. i, iv and v
- B. i, ii, iii
- C. i, iii, iv and v
- D. i, ii, iii, iv and v

Q4)[MAQ] Doraemon is calibrating his iconic Time Machine in space. The spacetime diagram from his 22nd-century gadget shows their time and space axes.

- Doraemon's Time Machine moves at $0.6c$ relative to Nobita.
- Gian's rocket moves at $0.88c$ relative to Nobita.
- Nobita is himself at rest at the origin.

The direction of motion of doreamon's time machine and gian's rocket travel relative to nobita are unknown.



Space for Rough Work

Which of the following statements are **CORRECT**?

- A. Nobita's rest frame corresponds to axes A (time) and B (space).
- B. A stationary event in Doraemon's frame will lie along a line parallel to axis C.
- C. If two events occur at the same location in Gian's frame, they lie along a line parallel to axis C.
- D. Gian's direction of motion is opposite to that of Doraemon.

5.[MAQ] Three inertial observers — Alice, Bob, and Charlie — witness two events, P and Q.

They report the following:

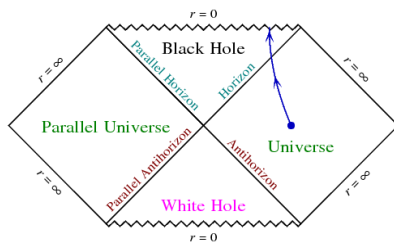
- Alice measures P and Q as space-like separated, with P occurring before Q.
- Bob also measures them as space-like separated, but Q occurs before P in his frame.
- Charlie measures P and Q as simultaneous in his frame.

(Assume a hypothetical faster-than-light signal is involved.)

Based on the principles of Special Relativity, which of the following are DIRECT CONSEQUENCES of these reports?

- A. There exists at least one inertial frame in which P and Q occur at the same time.
- B. There exists at least one inertial frame in which P and Q occur at the same location.
- C. No signal traveling at or below the speed of light could connect P and Q.
- D. A hypothetical FTL signal between the events could lead to a causal paradox.

6. A team of physicists places a perfectly reflecting mirror at a 45° angle inside a Schwarzschild black hole's event horizon. A beam of light from outside falls in, crosses the event horizon, and hits the mirror. Based on the causal structure shown in the Penrose diagram (with the mirror marked inside the horizon), what happens to the light after reflection?



- A. The light bounces back the way it came and escapes out of the black hole.
- B. The reflection splits the light into two beams — one going inward and one going backward in time — which cancel out.
- C. The light changes direction after reflection but still continues deeper inside the black hole, unable to escape back to the outside universe.
- D. The mirror sends the light so that it travels along the event horizon and remains stuck there forever.

Space for Rough Work

ANSWERS with hints

ENTANGLED SECRETS

- 1.A
- 2.B
- 3.D
- 4.C
- 5.A
- 6.B
- 7.C

SMALLER THAN YOU CAN SEE

- 1.B
- 2.A
- 3.A
- 4.A
- 5.C
- 6.C

ENGINES BENEATH

- 1.B
- 2.B
- 3.B
- 4.B
- 5.A
- 6.C

TAMING TIME AND SPACE

- 1.B
- 2.C
- 3.A
- 4.A,B,E
- 5.A,C,D
- 6.C

ENTANGLED SECRETS

1. A. Photon loss in space, weak entangled photon sources, need for quantum repeaters

Explanation: Photons can be lost due to scattering or absorption in space channels, and generating strong, stable entangled pairs is still difficult. Quantum repeaters are required to extend secure communication distances without losing entanglement.

2. B. It showed entanglement can survive across thousands of kilometers

Explanation: The Micius satellite experiment proved that quantum entanglement can be maintained over very large distances, confirming that quantum communication can work on a global scale.

3. D. Entanglement allows random mismatches by design

Explanation: Entanglement should produce perfect correlations when measured in the same basis. Random mismatches are caused by noise, detector errors, or eavesdropping, not by the nature of entanglement itself.

4. C. Measuring with the wrong basis destroys the key correlations

Explanation: In the BB84 protocol, the sender and receiver must use compatible measurement bases. If Bob's measurement system isn't aligned with Alice's photon states, the quantum correlations will be lost and no secure key can be formed.

5. A. Fiber optic cables scatter and absorb photons

Explanation: In urban environments, fiber losses, bending, and environmental interference degrade quantum signals. Photon absorption limits the distance and stability of quantum communication systems.

6. C. 33%

Explanation: There are 6 same-basis pairs, and 2 of them are errors (pairs 2 and 7).

Error rate = $2 \div 6 = 33\%$, indicating possible eavesdropping or channel noise.

7. B. Satellites maintain entanglement better because timing mismatches are smaller

Explanation: The smaller timing difference in the satellite link suggests better synchronization and less decoherence, helping maintain stronger entanglement correlations.

SMALLER THAN YOU CAN SEE

1. (ANSWER-B)

Explanation: Combustion rapidly consumes oxygen, creating low-oxygen conditions. Obligate anaerobes thrive most, while obligate aerobes suffer the most, giving the order III > IV > II > V > I.

2. (ANSWER-A)

Explanation:Electrolysis releases oxygen, increasing its concentration. Obligate aerobes benefit the most, while obligate anaerobes are harmed, giving $I > II > V > IV > III$.

3. (ANSWER-A)

Explanation:Without iodine, crystal violet is not fixed in any bacteria. Alcohol removes the dye from both types, and safranin stains both pink.

4. (ANSWER-A)

Explanation:Gold is dense, inert, and highly electron-scattering, producing sharp contrast. It is therefore ideal for antibody tagging in electron microscopy.

5. (ANSWER-C)

Explanation:The particle must be smaller than the distance between targets to avoid overlap. A 10 nm particle allows both proteins to be distinguished.

6. (ANSWER-C)

Explanation:Bacterial growth is exponential, doubling every 20 minutes. One quarter full occurs two doublings before full, which is 40 minutes earlier, at 5 hours 20 minutes after 10:00 AM.

ENGINES BENEATH

1. Ans: (B)

Explanation: Just increase d and decrease η . Blindly increasing ΔT can adversely affect the viscosity of the material, phase changes and plastic deformation, thus creating a counter effect.

2. Ans: B

Explanation:

The diffusion process is highly dependent on local concentrations and the randomness cannot be scaled to lead to an overall strong magnetic field, rather it will cancel out (more or less). Bulk motion can highly influence the field lines by stretching and twisting, each of which is a different concept altogether but very basic. Anyways, diffusion will die out eventually so obviously that's not how it works. Since the Reynolds number can be physically compared, it should be much greater than 1.

3. Ans: (B)

Explanation:

For a given k , (fourier's formula) larger q_{surf} (heat flux) implies larger temp gradient. Tectonically active zones will have larger q_{surf} because of increased exposure to mantle convection and everything follows. These zones have a thinner thermal boundary layer (ΔT in heat transfer mech para) and hence another reason for explosion of q_{surf} .

4. Ans (B)

Explanation: $n = P A x / Q f$

5. Ans: (A)

Explanation:

The fractional density difference is just $-\alpha \Delta T$ [let this be x]. $acceleration = g x$

6. Ans: C

Explanation:

Plate activity will decrease and will stop very abruptly (jump to 0 in very less time) Geological consequence: no plate movement, leading to thickening of lithosphere. No volcanic and mountain formation. Surface heat flux died because it's all just conduction now (for the heat still trapped). The magnetic field weakens and eventually dies. Surface temperature falls and the planet dies.

TAMING TIME AND SPACE

1. B(second image)

Explanation: as the slope of the worldline goes less than 1 which implies FTL.

2. C) The slope is 1.5, and F can be influenced by E.

Explanation: $Slope = ct/x = (3 \times 10^8) / (2 \times 10^8) = 1.5$

Angle $\theta = \arctan(1.5) \approx 56.3^\circ$

Light cone slope = 1 (45°). Since slope > 1 , F lies inside the causal reach of E (try drawing lightcones from event E to help understand).

3. A

Explanation: i)-Quantum entanglement does not involve the transfer of information so no violation of lightcone. iv)-not necessarily true that q has to lie

in the light cone of p. v)-there is no causal relation between measurements of Q and P.

4. 1,2,5

Explanation:

1 True – Nobita is at rest, so the standard perpendicular axes A (time) and B (space) represent his frame.

2 True – An object at rest in Doraemon's frame has a worldline parallel to Doraemon's time axis (C).

3 False – Line D does not match the required orientation of Gian's space axis.

4 False – Events at the same location in Gian's frame must be parallel to Gian's time axis, not axis C.

5 True – Doraemon's and Gian's time axes tilt in opposite directions, so they move in opposite directions.

5. A, C, D

Explanation:

A True For space-like separated events, there always exists an inertial frame in which the events are simultaneous. Charlie provides this frame.

B False If events are space-like separated, no inertial frame can make them occur at the same spatial location.

C True Space-like separation means no signal at or below the speed of light can causally connect P and Q.

D True An FTL signal could allow information to propagate backward in time in some frames, causing a causal paradox.

6. C

Explanation: Once light crosses the event horizon, all possible future paths lead further inward toward the singularity. A mirror inside the horizon can only change the direction of the light, not reverse it enough to let it escape. So even after bouncing off the mirror, the light is still trapped and continues deeper inside the black hole.